

[Classroom Transcript Targeted Mastery] Grade 6 Fortnight High-Frequency Vulnerability & Cognitive Blind Spot Comprehensive Exam (WS10)

Student Name:
Sophia / William

Time Allocated: 50 ~
60 Mins

Total Points:
100 Pts

Target Source:

Sept 12–20 Raw Transcripts + 53 Historical Cards

💡 Core Pedagogical Insights from Classroom Audio:

- ① **Dimensional Transformations:** When Volume is divided by 1D length, you MUST get a 2D perpendicular face; when gluing solids, the hidden contact face is subtracted TWICE!
- ② **Power Cycles & Prime Indices:** 2^n unit digits repeat in cycles of 4 (2, 4, 8, 6). For HCF/LCM in index form, select minimum/maximum exponents without manual expansion!
- ③ **Modulo Divisibility Strategy:** $1001 = 7 \times 11 \times 13 \implies 1000X \equiv -X \pmod{13}$. Fix unit digit Y first, then systematically test for valid single-digit solutions!

Part 1: Conceptual Precision & Mathematical Truth Statements

5 Questions, Total 20 Pts

1. [Audio 0912 · Divisibility by Powers of 10 Factors] Place Value Fundamentals (4 pts)

Teacher Quote: "Why do we only check the last two digits for 4 and 25? Because $100 = 4 \times 25$! What about 8 and 125?"

- (1) To determine whether an integer is divisible by 8 or 125, we only need to inspect its last _____ digits, because $1000 = \underline{\hspace{2cm}}$.
- (2) List all 3-digit positive multiples of 125 below 1000 (7 numbers in total):

2. [Audio 0912 · Primes & Composites Statements] Counterexamples & Logical Rigor (4 pts)

Determine whether each statement is True or False. If False, provide an explicit counterexample:

- (1) "All multiples of 5 are composite numbers." (Status: _____, Counterexample: _____)
- (2) "A composite number is always strictly greater than a prime number." (Status: _____, Counterexample: _____)

3. [Audio 0919 · Recurring Decimals & Statistics] Pure Cycles vs Range Definition (4 pts)

Teacher Quote: "Is the Mode the biggest number, or the most frequent? What does Range tell us about score consistency?"

- (1) Compare values: $1.2\dot{3}$ _____ $1.\dot{2}3$. (Fill in ">", "<", or "=")
- (2) In a class exam, the highest score was 98 and the lowest was 62. The score Range is _____. If 18 students scored 85 (the highest frequency), the Mode is _____.

4. [Audio 0919 · Geometric Dimensional Scaling] The Pizza & Cake Scaling Models (4 pts)

Teacher Quote: "A 12-inch pizza is sold out. The clerk offers two 6-inch pizzas instead. Do you take the deal? Why not?"

- (1) A "12-inch" pizza refers to its diameter. The area of a 12-inch pizza is _____ times that of a 6-inch pizza. Hence two 6-inch pizzas provide only _____ (fraction) of the original pizza!
- (2) A bakery sells an 8-inch cube cake. If replaced with 4-inch cube cakes, it takes exactly _____ smaller cakes to match the volume of one 8-inch cake.

5. [Audio 0912 · Nature of Exponents & Unit Digit Cycles] Periodicity Deduction (4 pts)

Teacher Quote: "What is the unit digit of 2^{2026} ? William said 6.5? 26 divided by 4 gives remainder 2, not 6.5!"

- (1) $2^4 = \underline{\hspace{2cm}}$. (Never confuse with $2 \times 4 = 8$!)
- (2) The unit digit of 2^n cycles every 4 steps: 2, 4, 8, 6. Since $2026 \div 4 = 506$ remainder $\underline{\hspace{1cm}}$, the unit digit of 2^{2026} is $\underline{\hspace{1cm}}$; similarly, the unit digit of 3^{2026} is $\underline{\hspace{1cm}}$.

Part 2: Dimensional Transformations & Metric Conversions

4 Questions, Total 20 Pts

6. [Audio 0919/0920 · Unit Conversion Traps] 1D, 2D, and 3D Metric Scaling (5 pts)

Rule: 1D length step = 10; 2D area step = 100; 3D volume step = 1000. Never omit units!

- (1) $7.04 \text{ m}^2 = \underline{\hspace{2cm}} \text{ dm}^2$ (2) $240 \text{ cm}^2 = \underline{\hspace{2cm}} \text{ dm}^2$
- (3) $1 \text{ m}^3 = \underline{\hspace{2cm}} \text{ dm}^3 = \underline{\hspace{2cm}} \text{ cm}^3$ (4) 6 tonnes 60 kg = $\underline{\hspace{2cm}}$ tonnes

7. [Audio 0920 · Dimensionality Reduction via Division] Spatial Physical Interpretation (5 pts)

Teacher Quote: "24 cm^3 divided by height 3 cm gives 8 cm^2 . Is 8 a perimeter, a length, or an area? Exactly which face?"

A cuboid has a volume of 24 cm^3 and a vertical height of 3 cm.

- (1) Calculate: $24 \text{ cm}^3 \div 3 \text{ cm} = \underline{\hspace{2cm}}$. (Include proper units!)
- (2) In terms of geometry and physical dimensions, explain why division causes "dimensionality reduction", and identify which faces of the cuboid this area represents (and why it cannot be the front or side faces):

Explanation:

8. [Audio 0919 · Scientific Notation & Order of Magnitude Checking] Fast Mental Verification (5 pts)

Teacher Quote: "Estimate 1.5×36 as $2 \times 30 = 60$. William got 5.4, which is off by an entire factor of 10!"

- (1) Mental multiplication: $240 \times 3.5 = \underline{\hspace{2cm}}$.
- (2) Use powers of 10 to evaluate $2,400,000 \times 0.000035$ cleanly without manual column counting:

$$2,400,000 \times 0.000035 = (240 \times 10\text{---}) \times (3.5 \times 10\text{---}) =$$

- (3) When calculating 1.5×36 , a student placed the decimal point incorrectly and wrote 5.4. Explain how mental estimation catches this error instantly:

Estimation Analysis:

9. [Audio 0919 · Distributive Law & Geometry Proof] Trapezoid Area Derivation (5 pts)

A diagonal divides a trapezoid with top base a , bottom base b , and height h into two triangles.

- (1) Area of lower triangle $S_1 = \underline{\hspace{2cm}}$; Area of upper triangle $S_2 = \underline{\hspace{2cm}}$.
- (2) Use factoring out the common factor $\frac{1}{2}h$ (reverse distributive property) to prove $S = \frac{1}{2}(a+b)h$:

Proof: $S = S_1 + S_2 =$

Part 3: Rigorous Calculation, Number Theory & Telescoping

4 Questions, Total 20 Pts

10. [Audio 0916/0920 · Mixed Numbers & True LCM] Careful Fraction Addition (5 pts)

Teacher Quote: "Sophia converted $3\frac{5}{6}$ into an improper fraction and wrote $18 + 5 = 90$! Slicing mooncakes requires true LCM!"

(1) Calculate: $2\frac{1}{3} + 3\frac{5}{6}$ (Show full working, avoiding the $18 + 5 = 90$ multiplication trap):

Working:

(2) Calculate: $\frac{1}{6} + \frac{1}{8}$ (Use $\text{LCM}(6, 8) = 24$; do not blindly multiply to 48):

Working:

11. [Audio 0912 · HCF and LCM in Index Form] Prime Power Extremum Rule (5 pts)

Teacher Quote: "Sophia saw $2^{10} \times 3^4$ and began multiplying out 1024 on scratch paper! Never expand; use prime power rules!"

Two positive integers are given in prime factorization form: $E = 2^{10} \times 3^4$, $F = 2^6 \times 3 \times 5^5$.

(1) Find $\text{HCF}(E, F)$ in prime factor index form: _____.

(2) Find $\text{LCM}(E, F)$ in prime factor index form: _____.

12. [Audio 0920 · Product Theorem: $a \times b = \text{GCD} \times \text{LCM}$] Factor Redundancy (5 pts)

Teacher Quote: "When we multiply 12 and 28 directly, how many times larger is it than their LCM? Exactly the GCD factor!"

Given positive integers 12 and 28:

(1) Use short division to find $\text{GCD}(12, 28) = \underline{\hspace{2cm}}$ and $\text{LCM}(12, 28) = \underline{\hspace{2cm}}$.

(2) Calculate: $12 \times 28 = \underline{\hspace{2cm}}$; $\text{GCD}(12, 28) \times \text{LCM}(12, 28) = \underline{\hspace{2cm}}$.

(3) Briefly explain why the product of any two positive integers always equals the product of their GCD and LCM:

Mathematical Rationale:

13. [Audio 0920 · Telescoping Series Summation] Fractional Difference Expansion (5 pts)

Teacher Quote: " $1/(1 \times 2) + 1/(2 \times 3) + \dots$ All intermediate terms cancel out, leaving only the first and last terms!"

(1) Prove the identity: $\frac{1}{n(n+1)} = \frac{1}{n} - \frac{1}{n+1}$ by expanding the right-hand side:

Proof: $\frac{1}{n} - \frac{1}{n+1} =$

(2) Evaluate the classroom summation using telescoping cancellation:

$$\frac{1}{1 \times 2} + \frac{1}{2 \times 3} + \frac{1}{3 \times 4} + \dots + \frac{1}{2025 \times 2026}$$

Working:

Final Value =

14. [Audio 0920 · Glued Cuboid & Cube Painted Surface Area] Double-overlap Deduction (10 pts)

Teacher Quote: "Glued $3 \times 4 \times 5$ cuboid and $2 \times 2 \times 2$ cube: how many overlapping faces are hidden? TWO! Sophia subtracted three times; William wrote 4D units!"

A large wooden cuboid measures $5 \text{ cm} \times 4 \text{ cm} \times 3 \text{ cm}$. A small wooden cube with edge length 2 cm is glued onto the center of one of its faces to form a composite solid. All exposed outer surfaces are to be spray-painted.

(1) Before gluing, the independent surface area of the cuboid is _____ cm^2 ; the cube is _____ cm^2 . (2 pts)

(2) When joined, the number of hidden overlapping faces is _____, with total hidden overlap area of _____ cm^2 . (2 pts)

(3) Calculate the total exposed surface area of the composite solid: (3 pts)

Calculation:

Ans: Exposed Surface Area = _____ cm^2 .

(4) If 1 cm^2 of surface requires 200 mg of paint, how many grams (g) of paint are needed in total? (3 pts)

Calculation:

Ans: Total paint needed = _____ grams.

15. [Audio 0920 · Three Stacking Configurations & School Homework Correction] (10 pts)

Teacher Quote: "The school teacher accidentally ticked 104 twice! William's stare method: smaller overlap gives bigger exposed area! 112 is the true maximum!"

Two identical wooden cuboids measuring $5 \text{ cm} \times 3 \text{ cm} \times 2 \text{ cm}$ are joined together face-to-face in three different configurations.

(1) Independent surface area of one cuboid is _____ cm^2 ; sum of two cuboids before gluing is _____ cm^2 . (2 pts)

(2) **Configuration A (glue 3×5 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(3) **Configuration B (glue 2×5 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(4) **Configuration C (glue 2×3 face):** Overlap area deducted = _____ cm^2 ; Resulting Surface Area = _____ cm^2 . (2 pts)

(5) In class, Sophia's school teacher marked both Config B and Config C as 104 cm^2 . Provide mathematical proof correcting this marking slip, and state the general inverse rule connecting overlap area and exposed surface area: (2 pts)

Proof & Inverse Law:

Part 5: Integrated Word Problems & Modulo Divisibility Finale 3 High-Order Problems, Total 20 Pts

16. [Audio 0916 · Speed, Distance, Time & Journey Fractions] (6 pts)

Teacher Quote: "William saw a word problem and left it completely blank! 1000m in 20 min; what fraction is 7 min?"

A runner completes a 1000 m distance on a circular track at a constant speed in 20 minutes.

(1) What is the runner's speed in meters per minute (m/min), and in kilometers per hour (km/h)? (3 pts)

Calculation:

Ans: Speed = _____ m/min = _____ km/h.

(2) What fraction of the total journey did the runner cover during the final 7 minutes? (3 pts)

Calculation:

Ans: Fraction of journey = _____.

17. [Audio 0916 · "Spending Money" Mental Model] Magnitude Ordering (6 pts)

Teacher Quote: "A - 2020 = B + 2021 = C - 2022 = D + 2023. Who spent the most and still matches the rest? C has the most money!"

Four distinct positive numbers A, B, C, D satisfy:

$$A - 2020 = B + 2021 = C - 2022 = D + 2023 = k \quad (k > 0)$$

(1) Use the intuitive "spending / receiving money" analogy to explain why C is the greatest and D is the least: (3 pts)

Intuitive Rationale:

(2) Express A, B, C, D algebraically in terms of k , and rank them from greatest to least using ">": (3 pts)

Expressions: $A =$ _____, $B =$ _____, $C =$ _____, $D =$ _____

Ordering: _____

18. [Audio 0916 · Grand Modulo 13 Divisibility Finale] 6-Digit Number Challenge (8 pts)

Teacher Quote: "91X93Y is divisible by 65. William stopped after $Y = 0, 5$! Use $1001 = 7 \times 11 \times 13 \pmod{13}$ logic to find X!"

A 6-digit number $\overline{91X93Y}$ (X, Y are single digits, $0 \leq X, Y \leq 9$) is divisible by 65.

(1) Since $65 = 5 \times 13$, the number must be divisible by both 5 and 13. The possible values for unit digit Y are _____. (1 pt)

(2) Knowing that $1001 = 7 \times 11 \times 13$ is a multiple of 13, prove that $1000X \equiv -X \pmod{13}$: (2 pts)

Proof: $1000X = 1001X - X =$

(3) Decompose $\overline{91X93Y} = 910000 + 1000X + 930 + Y$. Since $910000 = 13 \times 70000 \equiv 0 \pmod{13}$ and $930 = 71 \times 13 + 7 \equiv 7 \pmod{13}$, state the necessary and sufficient divisibility condition modulo 13: (_____) is divisible by 13. (2 pts)

(4) Complete the branch analysis to find all valid single-digit solutions: (3 pts)

Branch 1: If $Y = 0$, then:

Branch 2: If $Y = 5$, then:

Conclusion: Unique solution is $X =$ _____, $Y =$ _____; the 6-digit number is _____.

[Teacher Answer Key & Scoring Guide] (For Parent & Educator Reference)

Part 1: Conceptual Precision (20 pts)

- (1) 3; 8×125 (1 pt each); (2) 125, 250, 375, 500, 625, 750, 875 (2 pts for all 7).
- (1) False, counterexample: 5 is prime (2 pts); (2) False, counterexample: 4 is composite and $4 < 5$ (or $9 < 11$) (2 pts).
- (1) $>$ ($1.2333... > 1.2323...$, 2 pts); (2) Range = 36, Mode = 85 (1 pt each).
- (1) 4 times; $1/2$ (1 pt each); (2) 8 (2 pts, volume ratio is $(8/4)^3 = 8$).
- (1) 16 (2 pts); (2) Remainder 2, unit digit of 2^{2026} is 4, unit digit of 3^{2026} is 9 (2 pts).

Part 2: Dimensional Transformations & Units (20 pts)

- (1) 704; (2) 2.4; (3) 1000, 1,000,000; (4) 6.06 (1 pt each).
- (1) 8 cm^2 (1 pt); (2) Volume is $3D (L \times W \times H)$. Dividing by height H removes the vertical z-axis dimension, leaving the 2D horizontal plane ($L \times W$), which corresponds precisely to the top and bottom faces. The front and side faces both contain height H , so they cannot be represented once height is divided out (4 pts). Dividing by length 4cm yields 6 cm^2 (left/right side faces).
- (1) 840 (1 pt); (2) $240 \times 10^4 \times 3.5 \times 10^{-5} = 840 \times 10^{-1} = 84$ (2 pts); (3) Estimation: $1.5 \approx 2$, $36 \approx 30 \implies 2 \times 30 = 60$. The value 5.4 differs by an entire factor of 10, immediately exposing a misplaced decimal point (2 pts).
- (1) $\frac{1}{2}bh$, $\frac{1}{2}ah$ (1 pt each); (2) $S = \frac{1}{2}ah + \frac{1}{2}bh = \frac{1}{2}(a+b)h$ via factoring out $\frac{1}{2}h$ (3 pts).

Part 3: Rigorous Calculation & Number Theory (20 pts)

- (1) $2\frac{2}{6} + 3\frac{5}{6} = 5\frac{7}{6} = 6\frac{1}{6}$ or $\frac{37}{6}$ (3 pts); (2) $\frac{4}{24} + \frac{3}{24} = \frac{7}{24}$ (2 pts).
- (1) HCF = $2^6 \times 3$ (2 pts); (2) LCM = $2^{10} \times 3^4 \times 5^5$ (3 pts).
- (1) GCD = 4, LCM = 84 (1 pt each); (2) $12 \times 28 = 336$, $4 \times 84 = 336$ (1 pt each); (3) In direct multiplication, shared prime factors (GCD) are counted twice. Dividing by one copy of GCD yields the LCM, hence $a \times b = \text{GCD} \times \text{LCM}$ (1 pt).
- (1) $\frac{1}{n} - \frac{1}{n+1} = \frac{(n+1)-n}{n(n+1)} = \frac{1}{n(n+1)}$ (2 pts); (2) $(1 - \frac{1}{2}) + (\frac{1}{2} - \frac{1}{3}) + \dots + (\frac{1}{2025} - \frac{1}{2026}) = 1 - \frac{1}{2026} = \frac{2025}{2026}$ (3 pts).

Part 4: 3D Geometry & Surface Area (20 pts)

- (1) 94, 24 (1 pt each); (2) 2 hidden faces, overlap area = $2 \times 4 = 8 \text{ cm}^2$ (1 pt each); (3) $94 + 24 - 8 = 110 \text{ cm}^2$ (3 pts); (4) $110 \times 200 \text{ mg} = 22000 \text{ mg} = 22 \text{ g}$ (3 pts).
- (1) 62, 124 (1 pt each); (2) 30, 94 (1 pt each); (3) 20, 104 (1 pt each); (4) 12, 112 (1 pt each); (5) Config B overlap is $2 \times 10 = 20 \implies 104 \text{ cm}^2$; Config C overlap is $2 \times 6 = 12 \implies 112 \text{ cm}^2 \neq 104$. Law: smaller overlap area leaves larger exposed surface area (2 pts).

Part 5: Integrated Word Problems & Modulo Divisibility (20 pts)

- (1) Speed = $1000/20 = 50 \text{ m/min}$; $50 \times 60/1000 = 3 \text{ km/h}$ (3 pts); (2) $7/20$ (3 pts).
- (1) C spent the most money and still equals the others $\implies C$ started with the most; D had to receive the most to catch up $\implies D$ started with the least (3 pts); (2) $A = k + 2020$, $B = k - 2021$, $C = k + 2022$, $D = k - 2023 \implies C > A > B > D$ (3 pts).
- (1) $Y \in \{0, 5\}$ (1 pt); (2) $1000X = 1001X - X = 13 \times 77X - X \equiv -X \pmod{13}$ (2 pts); (3) Remainder 7; Divisibility requires $7 + Y - X \equiv 0 \pmod{13}$ (2 pts); (4) Branch 1 ($Y = 0$): $7 - X \equiv 0 \implies X = 7$. Branch 2 ($Y = 5$): $12 - X \equiv 0 \implies X = 12$ (invalid) or $12 - X = 13 \implies X = -1$ (invalid). Unique solution: $X = 7$, $Y = 0$, number is 917930 (3 pts).